

## 31338 Network Servers

### Lab 4b: Configuring a DHCP server on Linux

#### Aims

1. To configure a Linux machine as a DHCP server
2. To test the configuration by using a Windows client

#### Background Information

We will use the virtual switch, VMnet2, to create a small LAN shared by the Linux and Windows servers.

Initially, we will set up the Linux server as a DHCP server, with the Windows Server as a DHCP client  
Then we will do the opposite and use the Windows server as a DHCP server, with the Linux server as a DHCP client  
Both will run under a slightly different network configuration

Refer to Lab 4a on how to set up the machines as DHCP clients – we assume you know how to do this already.

#### Task 1: Install DHCP server on Linux

Use either yum or the GUI to install the standard **dhcp-server** on linux (e.g. `yum install dhcp-server`)

#### Task 2: Configure DHCP server on Linux

1. Design a subnet based on Lab 3a: We will use the 10.0.2.0/24 subnet with the default route going to the interface **ens37** (Ethernet 1). This interface will be allocated the IP address 10.0.2.1  
We need the following:

|                                                   |                         |
|---------------------------------------------------|-------------------------|
| Subnet:                                           | 10.0.2.0/24             |
| Gateway:                                          | 10.0.2.1                |
| DNS:                                              | 10.0.2.1                |
| Reserved space for servers:                       | 10.0.0.2 → 10.0.2.127   |
| Dynamically allocated space for workstations etc: | 10.0.2.128 → 10.0.2.254 |

(NOTE: Change the parameters to match your setup, e.g. your gateway might be different)

2. Read the sample `dhcpd.conf` file at:  
`/usr/share/doc/dhcp-server/dhcpd.conf.example`  
and then edit the **real** `/etc/dhcp/dhcpd.conf` file
3. Parameters to use:
  - set `default-lease-time` to 60 and `max-lease-time` to 600 (10 minutes)
  - create a subnet (10.0.2.0 netmask 255.255.255.0) with range 10.0.2.129-10.0.2.254
  - options `routers` 10.0.2.1; `domain-name-servers` 10.0.2.1, `domain-name` whatever.localdomain
  - The following is a sample:

```
subnet 10.0.2.0 netmask 255.255.255.0 {  
    range 10.0.2.128 10.0.2.254;  
    option domain-name-servers 10.0.2.1;  
    option domain-name "localdomain";  
}
```

4. Set up the DHCP server on ens37: We need to set up the ens37 interface to a static address 10.0.2.1 with mask 255.255.255.0 (recall Lab 3a? Hint:  
ifconfig ens37 to 10.0.2.1 ?? static (or can use GUI)  
(also edit the /etc/sysconfig/networking/devices/ifcfg-ens37 file to change  
BOOTPROTO=none)

### **Task 3: Start and monitor the DHCP server on Linux**

Use `systemctl` to **start** the `dhcpd` service. For now there is no need to enable it to start at boot.

One of the simplest ways to monitor the DHCP (and other servers) is to use the `-f` option on the `tail` command to watch entries being written to the main system log file (`/var/log/messages`):

```
tail -f /var/log/messages
```

Verify from the log file that the DHCP server is listening on the correct interface (ens37 only, not the other network interfaces).

You can also view DHCP database (leases issued):

```
/var/lib/dhcpd/dhcpd.leases
```

Note that there are no leases recorded in here at the moment...

### **Task 4: Set up Windows DHCP client & test the DHCP server**

1. Set up the Windows Server as a DHCP client (see Lab 4a)  
Hint: Basically you remove static network on Windows; set to automatic.  
(Network Connections → Ethernet1 → Properties → Internet Protocol Version 4 (TCP/IPv4) → Properties)
2. Refresh the network settings – this should be automatic, but sometimes you need to disable/enable the network interface to refresh the settings
3. You should see an IP address in the correct range and DNS and gateway settings correctly set  
(Hint: use the `netstat /all` command from a command prompt)

On the Linux DHCP server, check the following:

1. Check `/var/lib/dhcpd/dhcpd.leases` – you should see our windows machine hardware ethernet MAC address, IP address, Hostname and Start/End times.
2. Confirm our MAC address used on the ens37 interface by running:

```
arp -i ens37
```

You should also see the MAC address of the Windows Server machine.

3. View the log file (`/var/log/messages`) for DHCP messages. You should see the request/response sequence from the Windows Server, i.e.:
  - DHCPDISCOVER
  - DHCPOFFER
  - DHCPREQUEST
  - DHCPACK
4. You might also see a message from DHCPINFORM saying "not authoritative". Can you guess why?  
(Hint: if you modify `dhcpd.conf` and add the keyword "authoritative", you won't see this message)

***This might be a good time to make a backup of your working configuration files  
Also it is a good time to record all the above into your engineering journal.....***

### **Task 5: Set up reserved addresses**

Sometimes we need to allocate dedicated addresses for servers so they won't be reused by a user workstation. We call this a reserved address (or static address, or a reservation) and we need a key (usually the network adapter MAC or hardware Ethernet ID – which is a set of hexadecimal digits)

1. Our first step is to find out what our Windows Server's MAC address is.  
Hint: On Linux you should find it in `/var/lib/dhcpd/dhcpd.leases` or use the `arp` command
2. Next step: modify the DHCP server configuration file, i.e. modify `/etc/dhcp/dhcpd.conf` and add a host entry eg:

```
host WinServer {  
    hardware ethernet 00:0c:29:xx:yy:zz;  
    fixed-address 10.0.2.20;  
}
```

where 00:0c:29:xx:yy:zz is the MAC address of the “Ethernet1” adapter on the Windows Server machine.

3. Restart the DHCP server  
`systemctl restart dhcpd`
4. Testing Windows Server client:  
after one minute (60 seconds) should see the new static address appear on the Windows machine  
(recall: Use `ipconfig` command)  
Check & ping Linux again (10.0.2.1)
5. Check Linux logs  
On the Linux machine, check the lease again.  
Also try  
`arp -i ens37`  
What do you notice?  
What happens to the old lease?

***Don't forget to backup your files and configurations!***